

# Lab1 IP

## 2. A look at the captured trace

1. What is the IP address of your computer?

My IP address is: 192.168.68.60

| No.  | Time            | Source         | Destination    | Protocol | Length | Info                                                |
|------|-----------------|----------------|----------------|----------|--------|-----------------------------------------------------|
| 984  | 20:20:33.759194 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=1/256, ttl=1 (r  |
| 985  | 20:20:33.767900 | 192.168.68.1   | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 991  | 20:20:33.777214 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=2/512, ttl=1 (r  |
| 992  | 20:20:33.779808 | 192.168.68.1   | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 993  | 20:20:33.779948 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=3/768, ttl=1 (r  |
| 995  | 20:20:33.782465 | 192.168.68.1   | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 996  | 20:20:33.782563 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=4/1024, ttl=2 (  |
| 997  | 20:20:33.786942 | 192.168.0.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 1000 | 20:20:33.799586 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=5/1280, ttl=2 (  |
| 1001 | 20:20:33.805748 | 192.168.0.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 1002 | 20:20:33.803948 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=6/1536, ttl=2 (  |
| 1006 | 20:20:33.808238 | 192.168.0.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 1007 | 20:20:33.808369 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=7/1792, ttl=3 (  |
| 1008 | 20:20:33.829319 | 67.218.102.119 | 192.168.68.60  | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in tra |
| 1009 | 20:20:33.830799 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=8/2048, ttl=3 (  |
| 1010 | 20:20:33.854441 | 67.218.102.119 | 192.168.68.60  | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in tra |
| 1011 | 20:20:33.854614 | 192.168.68.60  | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=9/2304, ttl=3 (  |

2. Within the IP packet header, what is the value in the upper layer protocol field?

The value in the upper layer protocol field is ICMP(1) (Internet Control Message Protocol).

| No.  | Time            | Source          | Destination    | Protocol | Length | Info                                                |
|------|-----------------|-----------------|----------------|----------|--------|-----------------------------------------------------|
| 984  | 20:20:33.759194 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=1/256, ttl=1 (r  |
| 985  | 20:20:33.767900 | 192.168.68.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 991  | 20:20:33.777214 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=2/512, ttl=1 (r  |
| 992  | 20:20:33.779808 | 192.168.68.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 993  | 20:20:33.779948 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=3/768, ttl=1 (r  |
| 995  | 20:20:33.782465 | 192.168.68.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 996  | 20:20:33.782563 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=4/1024, ttl=2 (  |
| 997  | 20:20:33.786051 | 192.168.68.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 1000 | 20:20:33.799586 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=5/1280, ttl=2 (  |
| 1001 | 20:20:33.803748 | 192.168.68.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 1002 | 20:20:33.803948 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=6/1536, ttl=2 (  |
| 1006 | 20:20:33.808238 | 192.168.68.1    | 192.168.68.60  | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in tra |
| 1007 | 20:20:33.808369 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=7/1792, ttl=3 (  |
| 1008 | 20:20:33.829319 | 67.218.102.119  | 192.168.68.60  | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in tra |
| 1009 | 20:20:33.830799 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=8/2048, ttl=3 (  |
| 1010 | 20:20:33.854441 | 67.218.102.119  | 192.168.68.60  | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in tra |
| 1011 | 20:20:33.854614 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=9/2304, ttl=3 (  |
| 1012 | 20:20:33.875802 | 67.218.102.119  | 192.168.68.60  | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in tra |
| 1013 | 20:20:33.876131 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=10/2560, ttl=4 ( |
| 1014 | 20:20:33.891446 | 174.172.182.184 | 192.168.68.60  | ICMP     | 110    | Time-to-live exceeded (Time to live exceeded in tra |
| 1015 | 20:20:33.893251 | 192.168.68.60   | 128.119.245.12 | ICMP     | 70     | Echo (ping) request id=0xe472, seq=11/2816, ttl=4 ( |

Frame 984: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0  
Ethernet II, Src: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95), Dst: TPLink-9f:6a:ac (78:8c:b5:9f:6a:ac)

Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12

0100 ... = Version: 4  
... 0101 = Header Length: 20 bytes (5)  
~ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)  
0000 00.. = Differentiated Services Codepoint: Default (0)  
... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)  
Total Length: 56  
Identification: 0xe473 (58433)  
~ 000. .... = Flags: 0x0  
0... .... = Reserved bits: Not set  
0.. .... = Don't fragment: Not set  
..0. .... = More fragments: Not set  
...0 0000 0000 0000 = Fragment Offset: 0

~ Time to Live: 1  
Protocol: icmp (1)

Header Checksum: 0x5ae9 [validation disabled]  
[Header checksum status: Unverified]  
Source Address: 192.168.68.60  
Destination Address: 128.119.245.12  
[Stream index: 45]

Ethernet  
0 | | | |  
Internet Protocol  
0 | | | |  
Version 4

3. How many bytes are in the IP header? How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes.

There are 20 bytes in the IP header; The total length is 56 bytes, 20 bytes for the header, then only 36 bytes left is the payload of the IP datagram.

```
Unused: 00000000
v Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 56
  Identification: 0xddaf (56751)
  > 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
```

$$56 - 20 = 36$$

4. Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented.

This IP datagram is not been fragmented, because the fragment offset is 0 and it explicitly marked don't fragment: not set, and more fragments: not set.

```
v Internet Protocol Version 4, Src: 192.168.68.1, Dst: 192.168.68.60
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  v Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    1100 00.. = Differentiated Services Codepoint: Class Selector 6 (48)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 84
  Identification: 0xb376 (45942)
  v 000. .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: ICMP (1)
  Header Checksum: 0xbce4 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.68.1
  Destination Address: 192.168.68.60
```

5. Which fields in the IP datagram *a*lways change from one datagram to the next within this series of ICMP messages sent by your computer?

Fields in the IP datagram that always change are the header checksum and the identification.

| No.  | Time            | Source         | Destination   | Protocol | Length | Info                                                                            |
|------|-----------------|----------------|---------------|----------|--------|---------------------------------------------------------------------------------|
| 1747 | 20:21:13.744593 | 128.119.0.216  | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1749 | 20:21:13.833893 | 128.119.0.216  | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1751 | 20:21:13.987134 | 128.119.0.216  | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1944 | 20:21:29.378733 | 128.119.0.217  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1948 | 20:21:29.714230 | 128.119.0.217  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1950 | 20:21:29.825075 | 128.119.0.217  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 2092 | 20:21:45.971041 | 128.119.245.12 | 192.168.68.60 | ICMP     | 70     | Echo (ping) reply id=0xe472, seq=88/22528, ttl=31 (request exceeded in transit) |
| 2098 | 20:21:46.237917 | 128.119.245.12 | 192.168.68.60 | ICMP     | 70     | Echo (ping) reply id=0xe472, seq=89/22784, ttl=31 (request exceeded in transit) |
| 2100 | 20:21:46.330438 | 128.119.245.12 | 192.168.68.60 | ICMP     | 70     | Echo (ping) reply id=0xe472, seq=90/23040, ttl=31 (request exceeded in transit) |
| 1960 | 20:21:30.433398 | 128.119.3.32   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1964 | 20:21:30.762092 | 128.119.3.32   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1966 | 20:21:30.851173 | 128.119.3.32   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1952 | 20:21:29.921929 | 128.119.3.33   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1956 | 20:21:30.243176 | 128.119.3.33   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1958 | 20:21:30.334797 | 128.119.3.33   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1754 | 20:21:14.078520 | 128.119.77.233 | 192.168.68.60 | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1756 | 20:21:14.173657 | 128.119.77.233 | 192.168.68.60 | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1758 | 20:21:14.262196 | 128.119.77.233 | 192.168.68.60 | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1100 | 20:20:34.849875 | 163.253.1.156  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1102 | 20:20:34.946540 | 163.253.1.156  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1104 | 20:20:35.041869 | 163.253.1.156  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |

Frame 1950: 186 bytes on wire (1488 bits), 186 bytes captured (1488 bits) on interface en0, id 0

Ethernet II, Src: TPLink\_9f:6a:ac (78:8c:b5:9f:6a:ac), Dst: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95)

Internet Protocol Version 4, Src: 128.119.0.217, Dst: 192.168.68.60

0100 .... = Version: 4

.... 0101 = Header Length: 20 bytes (5)

✓ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

0000 00.. = Differentiated Services Codepoint: Default (0)

.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)

Total Length: 172

Identification: 0x5aac (23212)

✓ 010. .... = Flags: 0x2, Don't fragment

0... .... = Reserved bit: Not set

.1.. .... = Don't fragment: Set

..0. .... = More fragments: Not set

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 230

Protocol: ICMP (1)

Header Checksum: 0xb36f [validation disabled]

[Header checksum status: Unverified]

Source Address: 128.119.0.217

Destination Address: 192.168.68.60

[Stream index: 75]

Ethernet

Type IPv4

Internet Protocol Version 4

Version 4

Header 20

Identification 0x5aac (23212)

| No.  | Time            | Source         | Destination   | Protocol | Length | Info                                                                            |
|------|-----------------|----------------|---------------|----------|--------|---------------------------------------------------------------------------------|
| 1747 | 20:21:13.744593 | 128.119.0.216  | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1749 | 20:21:13.833893 | 128.119.0.216  | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1751 | 20:21:13.987134 | 128.119.0.216  | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1944 | 20:21:29.378733 | 128.119.0.217  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1948 | 20:21:29.714230 | 128.119.0.217  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1950 | 20:21:29.825075 | 128.119.0.217  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 2092 | 20:21:45.971041 | 128.119.245.12 | 192.168.68.60 | ICMP     | 70     | Echo (ping) reply id=0xe472, seq=88/22528, ttl=31 (request exceeded in transit) |
| 2098 | 20:21:46.237917 | 128.119.245.12 | 192.168.68.60 | ICMP     | 70     | Echo (ping) reply id=0xe472, seq=89/22784, ttl=31 (request exceeded in transit) |
| 2100 | 20:21:46.330438 | 128.119.245.12 | 192.168.68.60 | ICMP     | 70     | Echo (ping) reply id=0xe472, seq=90/23040, ttl=31 (request exceeded in transit) |
| 1960 | 20:21:30.433398 | 128.119.3.32   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1964 | 20:21:30.762092 | 128.119.3.32   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1966 | 20:21:30.851173 | 128.119.3.32   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1952 | 20:21:29.921929 | 128.119.3.33   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1956 | 20:21:30.243176 | 128.119.3.33   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1958 | 20:21:30.334797 | 128.119.3.33   | 192.168.68.60 | ICMP     | 70     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1754 | 20:21:14.078520 | 128.119.77.233 | 192.168.68.60 | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1756 | 20:21:14.173657 | 128.119.77.233 | 192.168.68.60 | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1758 | 20:21:14.262196 | 128.119.77.233 | 192.168.68.60 | ICMP     | 98     | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1100 | 20:20:34.849875 | 163.253.1.156  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1102 | 20:20:34.946540 | 163.253.1.156  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |
| 1104 | 20:20:35.041869 | 163.253.1.156  | 192.168.68.60 | ICMP     | 186    | Time-to-live exceeded (Time to live exceeded in transit)                        |

Frame 2092: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0

Ethernet II, Src: TPLink\_9f:6a:ac (78:8c:b5:9f:6a:ac), Dst: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95)

Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.68.60

0100 .... = Version: 4

.... 0101 = Header Length: 20 bytes (5)

✓ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

0000 00.. = Differentiated Services Codepoint: Default (0)

.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)

Total Length: 56

Identification: 0x52bb (21179)

✓ 000. .... = Flags: 0x0

0... .... = Reserved bit: Not set

0... .... = Don't fragment: Not set

..0. .... = More fragments: Not set

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 31

Protocol: ICMP (1)

Header Checksum: 0xcea1 [validation disabled]

[Header checksum status: Unverified]

Source Address: 128.119.245.12

Destination Address: 192.168.68.60

[Stream index: 45]

Internet Control Message Protocol

Ethernet

Version 4

Time

6. Which fields stay constant? Which of the fields *must* stay constant? Which fields must change? Why?

Fields that stay constant are: IP version(IPv4), length of header(20 bytes), SourceIP(sending from the same palace), Destination IP(contacting same site), Upper layer protocol is always ICMP(1);

The fields must stay constant, the same field must stay constant;

The fields that must change are: identification, and checksum(header & ICMP). The identification needs to change because each new packet gets a new unique ID to help reassemble fragments if needed. The IP Header Checksum changes since the TTL and other fields modify the header. The ICMP Checksum changes due to variations in the encapsulated IP header.

7. Describe the pattern you see in the values in the Identification field of the IP datagram

The pattern in the identification field is that the field increases by one in each strand of echo requests

|      |                 |               |                |      |    |                     |                |
|------|-----------------|---------------|----------------|------|----|---------------------|----------------|
| 984  | 20:20:33.759194 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 991  | 20:20:33.777214 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 993  | 20:20:33.779948 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 996  | 20:20:33.782563 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 1000 | 20:20:33.799586 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 1002 | 20:20:33.803948 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 1007 | 20:20:33.808369 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 1009 | 20:20:33.830799 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 1011 | 20:20:33.854614 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 1013 | 20:20:33.876131 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |
| 1015 | 20:20:33.897751 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq |

|                                                                                                   |
|---------------------------------------------------------------------------------------------------|
| Frame 984: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0       |
| Ethernet II, Src: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac) |
| Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12                              |
| 0100 .... = Version: 4                                                                            |
| .... 0101 = Header Length: 20 bytes (5)                                                           |
| ✓ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)                                   |
| 0000 00.. = Differentiated Services Codepoint: Default (0)                                        |
| .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)                       |
| Total Length: 56                                                                                  |
| Identification: 0xe473 (58483)                                                                    |

1,

+1 = 58484

|      |                 |               |                |      |    |                     |                   |
|------|-----------------|---------------|----------------|------|----|---------------------|-------------------|
| 993  | 20:20:33.779948 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=3  |
| 996  | 20:20:33.782563 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=4  |
| 1000 | 20:20:33.799586 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=5  |
| 1002 | 20:20:33.803948 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=6  |
| 1007 | 20:20:33.808369 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=7  |
| 1009 | 20:20:33.830799 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=8  |
| 1011 | 20:20:33.854614 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=9  |
| 1013 | 20:20:33.876131 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=10 |
| 1015 | 20:20:33.893251 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=11 |
| 1017 | 20:20:33.918058 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=12 |

Frame 993: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0  
 Ethernet II, Src: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95), Dst: TPLink\_9f:6a:ac (78:8c:b5:9f:6a:ac)  
 Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12

```

0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 56
Identification: 0xe475 (58485)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
..0. .... = Don't fragment: Not set

```

|      |                 |               |                |      |    |                                                          |                                             |
|------|-----------------|---------------|----------------|------|----|----------------------------------------------------------|---------------------------------------------|
| 995  | 20:20:33.782465 | 192.168.68.1  | 192.168.68.60  | ICMP | 98 | Time-to-live exceeded (Time to live exceeded in transit) |                                             |
| 984  | 20:20:33.759194 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=1/256, ttl=1 (no response)   |
| 991  | 20:20:33.777214 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=2/512, ttl=1 (no response)   |
| 993  | 20:20:33.779948 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=3/768, ttl=1 (no response)   |
| 996  | 20:20:33.782563 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=4/1024, ttl=2 (no response)  |
| 1000 | 20:20:33.799586 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=5/1280, ttl=2 (no response)  |
| 1002 | 20:20:33.803948 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=6/1536, ttl=2 (no response)  |
| 1007 | 20:20:33.808369 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=7/1792, ttl=3 (no response)  |
| 1009 | 20:20:33.830799 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=8/2048, ttl=3 (no response)  |
| 1011 | 20:20:33.854614 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=9/2304, ttl=3 (no response)  |
| 1013 | 20:20:33.876131 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request                                      | id=0xe472, seq=10/2560, ttl=4 (no response) |

Frame 991: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0  
 Ethernet II, Src: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95), Dst: TPLink\_9f:6a:ac (78:8c:b5:9f:6a:ac)  
 Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12

```

0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 56
Identification: 0xe474 (58484)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
..0. .... = Don't fragment: Not set
...0 .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0

```

$$58484 + 1 = 58485$$

8. What is the value in the Identification field and the TTL field?

Identification field is: 0Xe473(58483)

TTL field is: 1

|      |                 |               |                |      |    |                     |                                            |
|------|-----------------|---------------|----------------|------|----|---------------------|--------------------------------------------|
| 984  | 20:20:33.759194 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=1/256, ttl=1 (no response)  |
| 991  | 20:20:33.777214 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=2/512, ttl=1 (no response)  |
| 993  | 20:20:33.779948 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=3/768, ttl=1 (no response)  |
| 996  | 20:20:33.782563 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=4/1024, ttl=2 (no response) |
| 1000 | 20:20:33.799586 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=5/1280, ttl=2 (no response) |
| 1002 | 20:20:33.803948 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=6/1536, ttl=2 (no response) |
| 1007 | 20:20:33.808369 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=7/1792, ttl=3 (no response) |
| 1009 | 20:20:33.830799 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=8/2048, ttl=3 (no response) |
| 1011 | 20:20:33.854614 | 192.168.68.60 | 128.119.245.12 | ICMP | 70 | Echo (ping) request | id=0xe472, seq=9/2304, ttl=3 (no response) |

Frame 984: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0  
 Ethernet II, Src: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95), Dst: TPLink\_9f:6a:ac (78:8c:b5:9f:6a:ac)  
 Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12

```

0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 56
Identification: 0xe473 (58483)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
..0. .... = Don't fragment: Not set
...0 .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0

```

```

Time to Live: 1
Protocol: ICMP (1)
Header Checksum: 0x5ae9 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.68.60
Destination Address: 128.119.245.12

```

9. Do these values remain unchanged for all of the ICMP TTL-exceeded replies sent to your computer by the nearest (first hop) router? Why?

For the identification field changes with each packet because each ICMP reply must have a unique ID, if multiple replies have the same ID, they would be treated as fragments of a larger packet.

The TTL field does not change because the time to live to the first hop router is always the same.

```
997 20:20:33.786942 192.168.0.1 192.168.68.60 ICMP 98 Time-to-live excees
1001 20:20:33.803748 192.168.0.1 192.168.68.60 ICMP 98 Time-to-live excees
1006 20:20:33.808238 192.168.0.1 192.168.68.60 ICMP 98 Time-to-live excees
985 20:20:33.767900 192.168.68.1 192.168.68.60 ICMP 98 Time-to-live excees
992 20:20:33.779808 192.168.68.1 192.168.68.60 ICMP 98 Time-to-live excees
995 20:20:33.782465 192.168.68.1 192.168.68.60 ICMP 98 Time-to-live excees
984 20:20:33.759194 192.168.68.60 128.119.245.12 ICMP 70 Echo (ping) reques
991 20:20:33.777214 192.168.68.60 128.119.245.12 ICMP 70 Echo (ping) reques
993 20:20:33.779948 192.168.68.60 128.119.245.12 ICMP 70 Echo (ping) reques
996 20:20:33.782563 192.168.68.60 128.119.245.12 ICMP 70 Echo (ping) reques

> Frame 997: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
> Ethernet II, Src: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac), Dst: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95)
> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.68.60
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    1100 00.. = Differentiated Services Codepoint: Class Selector 6 (48)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 84
  Identification: 0x747b (29819)
  > 000. .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 63
  Protocol: ICMP (1)
  Header Checksum: 0x40e0 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.1
  Destination Address: 192.168.68.60
```

no change, 63

```
997 20:20:33.786942 192.168.0.1 192.168.68.60 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
1001 20:20:33.803748 192.168.0.1 192.168.68.60 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
1006 20:20:33.808238 192.168.0.1 192.168.68.60 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
985 20:20:33.767900 192.168.68.1 192.168.68.60 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
992 20:20:33.779808 192.168.68.1 192.168.68.60 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
995 20:20:33.782465 192.168.68.1 192.168.68.60 ICMP 98 Time-to-live exceeded (Time to live exceeded in transit)
984 20:20:33.759194 192.168.68.60 128.119.245.12 ICMP 70 Echo (ping) request id=0xe472, seq=1/256, ttl=1 (no res
991 20:20:33.777214 192.168.68.60 128.119.245.12 ICMP 70 Echo (ping) request id=0xe472, seq=2/512, ttl=1 (no res
993 20:20:33.779948 192.168.68.60 128.119.245.12 ICMP 70 Echo (ping) request id=0xe472, seq=3/768, ttl=1 (no res
996 20:20:33.782563 192.168.68.60 128.119.245.12 ICMP 70 Echo (ping) request id=0xe472, seq=4/1024, ttl=2 (no re

> Frame 1001: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
> Ethernet II, Src: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac), Dst: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95)
> Internet Protocol Version 4, Src: 192.168.0.1, Dst: 192.168.68.60
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
    1100 00.. = Differentiated Services Codepoint: Class Selector 6 (48)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 84
  Identification: 0x7486 (29830)
  > 000. .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 63
  Protocol: ICMP (1)
  Header Checksum: 0x40d5 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.1
  Destination Address: 192.168.68.60
```

|                         |            |
|-------------------------|------------|
| Ethernet                |            |
| 0                       |            |
| Internet Protocol Versi |            |
| 0                       |            |
| Version                 | Header ... |
| 4                       | 20         |
| Identi                  |            |
| 0x7486                  |            |

## Fragmentation

10 Find the first ICMP Echo Request message that was sent by your computer after you changed the *Packet Size* in *pingplotter* to be 2000. Has that message been fragmented across more than one IP datagram? [Note: if you find your packet has not been fragmented, you should download the zip file

<http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces.zip> and extract the *ip-ethereal-trace-1* packet trace. If your computer has an Ethernet interface, a packet size of 2000 *should* cause fragmentation.<sup>3]</sup>

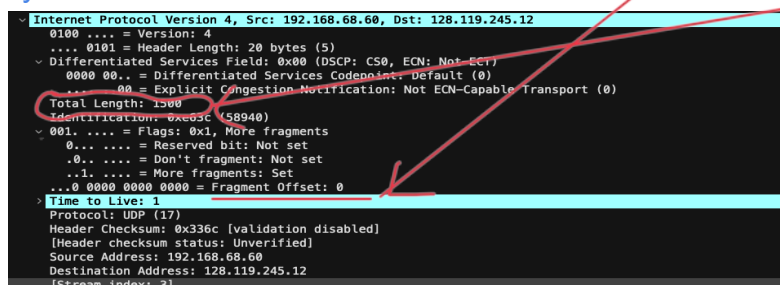
Yes, the message has been fragmented across more than one IP datagram according to the provided packet trace.



```
Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 1500
  Identification: 0xe03c (58940)
  001. .... = Flags: 0x1, More fragments
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..1. .... = More fragments: Set
  ...0 0000 0000 0000 = Fragment Offset: 0
  > Time to Live: 1
  Protocol: UDP (17)
  Header Checksum: 0x336c [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.68.60
  Destination Address: 128.119.245.12
  [Stream index: 3]
```

11. Print out the first fragment of the fragmented IP datagram. What information in the IP header indicates that the datagram been fragmented? What information in the IP header indicates whether this is the first fragment versus a latter fragment? How long is this IP datagram?

The flag indicating "more fragments" is set, confirming that the datagram has been fragmented. The fragment offset is 0, meaning this is the first fragment rather than a later one, where the offset would be 1480. The total length of the datagram is 1500 bytes.



```
Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 1500
  Identification: 0xe03c (58940)
  001. .... = Flags: 0x1, More fragments
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..1. .... = More fragments: Set
  ...0 0000 0000 0000 = Fragment Offset: 0
  > Time to Live: 1
  Protocol: UDP (17)
  Header Checksum: 0x336c [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.68.60
  Destination Address: 128.119.245.12
  [Stream index: 3]
```

12. Print out the second fragment of the fragmented IP datagram. What information in the IP header indicates that this is not the first datagram fragment? Are there more fragments? How can you tell?

The second fragment is obvious because it now has a fragment offset of 1480. There are no more fragments because it no longer has a flag set for more fragments

```
745 21:35:31.682712 128.119.0.216 192.168.68.60 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
747 21:35:31.683234 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=70/17920, ttl=24 (no response found!)
748 21:35:31.779501 128.119.77.233 192.168.68.60 ICMP 590 Time-to-live exceeded (Time to live exceeded in transit)
750 21:35:31.781290 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=71/18176, ttl=24 (no response found!)
751 21:35:31.910744 128.119.77.233 192.168.68.60 ICMP 590 Time-to-live exceeded (Time to live exceeded in transit)
753 21:35:31.911310 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=72/18432, ttl=24 (no response found!)
754 21:35:32.006339 128.119.77.233 192.168.68.60 ICMP 590 Time-to-live exceeded (Time to live exceeded in transit)
756 21:35:32.006866 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=73/18688, ttl=25 (no response found!)
861 21:35:37.012022 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=74/18944, ttl=25 (no response found!)
967 21:35:42.017126 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=75/19200, ttl=25 (no response found!)
1049 21:35:47.018997 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=76/19456, ttl=26 (no response found!)
1062 21:35:47.178267 128.119.0.217 192.168.68.60 ICMP 186 Time-to-live exceeded (Time to live exceeded in transit)
1064 21:35:47.179240 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=77/19712, ttl=26 (no response found!)
1065 21:35:47.274716 128.119.0.217 192.168.68.60 ICMP 186 Time-to-live exceeded (Time to live exceeded in transit)
1067 21:35:47.275202 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=78/19968, ttl=26 (no response found!)
1068 21:35:47.372381 128.119.0.217 192.168.68.60 ICMP 186 Time-to-live exceeded (Time to live exceeded in transit)
1070 21:35:47.372928 192.168.68.60 128.119.245.12 ICMP 534 Echo (ping) request id=0xe738, seq=79/20224, ttl=27 (no response found!)
1071 21:35:47.468604 128.119.3.33 192.168.68.60 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
```

```
> Frame 1546: 534 bytes on wire (4272 bits), 534 bytes captured (4272 bits) on interface en0, id 0
> Ethernet II, Src: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac), Dst: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95)
> Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.68.60
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 520
  Identification: 0x2f99 (12185)
  > 0000 .... = Flags: 0x0
  ...0 0000 1011 1001 = Fragment Offset: 1480
  Time to Live: 31
  Protocol: ICMP (1)
  Header Checksum: 0xef3a [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 128.119.245.12
  Destination Address: 192.168.68.60
  > [2 IPv4 Fragments (1980 bytes): #1545(1480), #1546(500)]
  [Stream index: 5]
> Internet Control Message Protocol
  Type: 0 (Echo (ping) reply)
  Code: 0
  Checksum: 0x186e [correct]
  [Checksum Status: Good]
  Identifier (OE): 50192 (0xe738)
  Identifier (LE): 14567 (0x3be7)
  Sequence Number (BE): 89 (0x8059)
```

13. What fields change in the IP header between the first and second fragment?

The fields that change are length, flags set, fragment offset, and header checksum.

Now find the first ICMP Echo Request message that was sent by your computer after you changed the *Packet Size* in *pingplotter* to be 3500.

14. How many fragments were created from the original datagram?

After switching to 3500 bytes, 3 fragments are created.

```
2762 21:41:21.199157 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=78/19968, ttl=26 (no response found!)
2781 21:41:21.307090 128.119.0.217 192.168.68.60 ICMP 186 Time-to-live exceeded (Time to live exceeded in transit)
2785 21:41:21.307556 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=79/20224, ttl=27 (no response found!)
2788 21:41:21.574300 128.119.3.32 192.168.68.60 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2800 21:41:21.750967 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=80/20480, ttl=27 (no response found!)
2817 21:41:22.100544 128.119.3.32 192.168.68.60 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2830 21:41:22.101274 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=81/20736, ttl=27 (no response found!)
2835 21:41:22.211800 128.119.3.32 192.168.68.60 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2839 21:41:22.212221 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=82/20992, ttl=28 (no response found!)
2841 21:41:22.324000 128.119.3.32 192.168.68.60 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2849 21:41:22.624971 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=83/21248, ttl=28 (no response found!)
2876 21:41:22.718450 128.119.3.32 192.168.68.60 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2879 21:41:22.718979 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=84/21504, ttl=28 (no response found!)
2899 21:41:22.833120 128.119.3.32 192.168.68.60 ICMP 70 Time-to-live exceeded (Time to live exceeded in transit)
2904 21:41:22.833707 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=85/21760, ttl=29 (no response found!)
3078 21:41:27.830796 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=86/22016, ttl=29 (no response found!)
3144 21:41:32.844055 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=87/22272, ttl=29 (no response found!)
3171 21:41:37.847338 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=88/22528, ttl=30 (reply in 3174)
3174 21:41:37.944242 128.119.245.12 192.168.68.60 ICMP 554 Echo (ping) reply id=0xe776, seq=88/22528, ttl=31 (request in 3171)
3180 21:41:38.114815 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=89/22784, ttl=30 (reply in 3183)
3183 21:41:38.217845 128.119.245.12 192.168.68.60 ICMP 554 Echo (ping) reply id=0xe776, seq=89/22784, ttl=31 (request in 3180)
3186 21:41:38.218162 192.168.68.60 128.119.245.12 ICMP 554 Echo (ping) request id=0xe776, seq=90/23040, ttl=30 (reply in 3189)
3189 21:41:38.344374 128.119.245.12 192.168.68.60 ICMP 554 Echo (ping) reply id=0xe776, seq=90/23040, ttl=31 (request in 3186)

> Frame 164: 554 bytes on wire (4432 bits), 554 bytes captured (4432 bits) on interface en0, id 0
> Ethernet II, Src: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)
> Internet Protocol Version 4, Src: 192.168.68.60, Dst: 128.119.245.12
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 540
  Identification: 0xe777 (59255)
  > 0000 .... = Flags: 0x0
  ...0 0001 0111 0010 = Fragment Offset: 2960
  > Time to Live: 1
  Protocol: ICMP (1)
  Header Checksum: 0x548f [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.68.60
  Destination Address: 128.119.245.12
  > [3 IPv4 Fragments (3480 bytes): #162(1480), #163(1480), #164(520)]
    [Frame: 162, payload: 0-1479 (1480 bytes)]
    [Frame: 163, payload: 1480-2959 (1480 bytes)]
    [Frame: 164, payload: 2960-3479 (520 bytes)]
    [fragment count: 3]
  [Reassembled IPv4 length: 3480]
```

15. What fields change in the IP header among the fragments?

The fields that change are the fragment offset and checksum. The first 2 packets also have lengths of 1480 and more fragments flags set, while the last fragment is shorter (520) and does not have a flag set.